

PROBLEM STATEMENT

Radical Exponential & Chain Rule

Find the derivative with respect to x of the function:

$$y = \sqrt{e^x + 1}$$

STEP 1 প্রদত্ত সূচকীয় বর্গমূল ফাংশনটি ঘাত আকারে প্রকাশ করি (Expressing in Exponent Power Form)

$$y = (e^x + 1)^{1/2}$$

STEP 2 উভয়পক্ষে x -এর সাপেক্ষে অন্তরীকরণ অপারেটর প্রয়োগ করি (Applying Differentiation Operator w.r.t. x)

$$\frac{dy}{dx} = \frac{d}{dx} [(e^x + 1)^{1/2}]$$

STEP 3 বহিঃস্থ সূচকের জন্য শক্তি বিধি (Power Rule) প্রয়োগ করি (Applying Power Rule to Outer Exponent)

$$\frac{dy}{dx} = \frac{1}{2}(e^x + 1)^{\frac{1}{2}-1} \cdot \frac{d}{dx}(e^x + 1) = \frac{1}{2}(e^x + 1)^{-1/2} \cdot \frac{d}{dx}(e^x + 1)$$

STEP 4 অভ্যন্তরীণ সূচকীয় পদের অন্তরজ নির্ণয় করি (Differentiating Inner Exponential Expression)

$$\frac{d}{dx}(e^x + 1) = \frac{d}{dx}(e^x) + \frac{d}{dx}(1) = e^x + 0 = e^x$$

STEP 5 মান প্রতিস্থাপন করে চূড়ান্ত সরল ভগ্নাংশ পাই (Substituting Derivative & Writing in Standard Radical Form)

$$\frac{dy}{dx} = \frac{1}{2\sqrt{e^x + 1}} \cdot e^x = \frac{e^x}{2\sqrt{e^x + 1}}$$

FINAL ANSWER

$$\frac{dy}{dx} = \frac{e^x}{2\sqrt{e^x + 1}}$$

 BOX 1: MATHEMATICAL FOUNDATIONS & FORMULA REFERENCE

- **Standard Radical Derivative:** $\frac{d}{dx}[\sqrt{u}] = \frac{1}{2\sqrt{u}} \frac{du}{dx}$.
- **Natural Exponential Derivative:** $\frac{d}{dx}[e^x] = e^x$.
- **Chain Rule Chain Composition:** With $u = e^x + 1$, $\frac{dy}{dx} = \frac{dy}{du} \frac{du}{dx} = \frac{1}{2\sqrt{u}} \cdot e^x$.

 BOX 2: COMMON STUDENT TRAPS & PITFALLS

- **Radical Distribution Fallacy:** Assuming $\sqrt{e^x + 1} = \sqrt{e^x} + \sqrt{1} = e^{x/2} + 1$. Square roots never distribute over sums!
- **Missing Constant Derivative:** Correctly recognising $\frac{d}{dx}(1) = 0$, leaving solely e^x from the inner term.
- **Missing Coefficient $\frac{1}{2}$:** Dropping the factor of 2 in the denominator during algebraic formatting.

 BOX 3: REAL DOMAIN & DIFFERENTIABILITY ANALYSIS

- **Domain of Definition:** Since $e^x > 0$ for all real x , we have $e^x + 1 > 1 > 0$. The expression under the radical is strictly positive everywhere.
- **Strict Real Domain:** Domain = $(-\infty, \infty) = \mathbb{R}$.
- **Smoothness:** Infinitely differentiable (C^∞) everywhere on the real line.

 BOX 4: LOGARITHMIC DIFFERENTIATION ALTERNATIVE

- Taking natural log: $\ln y = \frac{1}{2} \ln(e^x + 1)$
- Differentiating implicitly:

$$\frac{1}{y} \frac{dy}{dx} = \frac{1}{2} \cdot \frac{e^x}{e^x + 1}$$

- Multiplying by $y = \sqrt{e^x + 1}$:

$$\frac{dy}{dx} = \sqrt{e^x + 1} \cdot \frac{e^x}{2(e^x + 1)} = \frac{e^x}{2\sqrt{e^x + 1}}$$

 BOX 5: EXAM MARKS BREAKDOWN (AMIE SECTION-A FORMAT)

- **Step 1-2 (Power Formulation & Setup):** 1.0 Mark
- **Step 3 (Power Rule Outer Derivative):** 1.0 Mark
- **Step 4 (Exponential Inner Derivative):** 1.0 Mark
- **Step 5 & Final Answer:** 1.0 Mark
- **Total Allocation: 4.0 Marks**